

Product Requirements – Computer-Use Automation System

Overview

A backend system that lets AI agents operate legacy web applications with no API by recording an LLM-driven run as a reusable capability artifact and replaying it deterministically in production.

User Journeys

Capture: Human types a natural language goal → LLM navigates live browser → artifact saved.

Replay: Tool calls replay API with inputs → system executes without LLM → returns typed outputs or structured error.

Escalation: Automation stuck → intervention request sent → Human takes over live browser → signals done → automation resumes.

Functional Requirements

Capture: Accept a natural language goal and target application. Run an LLM-driven observe → decide → act loop until goal is met or stopping condition is hit. Extract stable element signals at each step. Save a structured, versioned artifact on completion.

Artifact: Capture ordered steps, ranked element targeting strategies, typed inputs/outputs, checkpoints, known business outcomes, and recoverable conditions per step. Tenant-scoped with optional base artifact inheritance. Must never contain credentials or PII.

Deterministic Replay: Re-execute without LLM. Locate elements via stored strategies in priority order. Classify failures as business outcome, recoverable condition, or hard failure. Return typed outputs on success.

Human Escalation: Detect when automation cannot proceed and pause. Emit intervention request with goal, step, reason, and screenshot. Expose the same live browser for human control. On signal: preserve session state and resume. Record intervention with before/after screenshot and notes.

Safety: Enforce per-artifact domain allowlist. Classify actions as read-only, write, or destructive — block destructive actions. Never persist credentials or PII.

Non-Functional Requirements

Requirement	Description
Determinism	Same artifact + inputs = same result on a stable UI
Recoverability	Resumes after human handoff with no session loss
Auditability	Every run produces a structured log sufficient to debug any failure
Security	No secrets in artifacts or logs. Credentials via environment variables only
Extensibility	Schema and replay engine support desktop surfaces without changes
Tenant isolation	Artifacts scoped to tenants. Base artifacts inherited but not mutated

Limitations

- Web surfaces only – desktop support designed but not built
- Multi-tenant infrastructure deferred – schema supports it, routing not built
- Capture run required before replay is available